

HDOC DAY-30

Title: Array Manipulation: Bubble Sort, Binary Search, Insertion, Deletion, and Right Rotation

1. Problem Statement Write a C program to perform the following menu-driven operations on a given array:

1. Sort elements using Bubble Sort algorithm and perform Binary Search.
2. Perform Insertion and Deletion in a given valid position.
3. Rotate array elements to the right by k positions.
4. Exit.

2. Algorithms

1. Bubble Sort and Binary Search Algorithm

- **Bubble Sort:** Loop i from 0 to $n - 2$; loop j from 0 to $n - i - 2$. Swap adjacent elements if $\text{arr}[j] > \text{arr}[j+1]$.
- **Binary Search:** Set $\text{low} = 0$, $\text{high} = n - 1$. While $\text{low} \leq \text{high}$, calculate $\text{mid} = \text{low} + (\text{high} - \text{low})/2$. If $\text{arr}[\text{mid}] == \text{key}$, return position; if $\text{arr}[\text{mid}] < \text{key}$, set $\text{low} = \text{mid} + 1$; else set $\text{high} = \text{mid} - 1$.

2. Insertion and Deletion Algorithm

- **Insertion:** Validate position $p \in [1, n + 1]$. Shift elements from index $n - 1$ down to $p - 1$ right by 1 position. Place new value at $\text{arr}[p-1]$ and increment n .
- **Deletion:** Validate position $p \in [1, n]$. Shift elements from index p up to $n - 1$ left by 1 position. Decrement n .

3. Right Rotation Algorithm

- Read k and adjust $k = k \pmod{n}$.
- Create temporary array $\text{rotated}[n]$. Assign $\text{rotated}[(i + k) \% n] = \text{arr}[i]$ for all $i \in [0, n - 1]$.

3. Test Case Table

Test Case	Option	Input Data	Expected Output	Actual Output	Status
1	Sort & Search	$N = 5$, [40, 10, 50, 20, 30], Search	Sorted: [10, 20, 30, 40, 50], Key found at index 1	Matched expected output	Pass

Test Case	Option	Input Data	Expected Output	Actual Output	Status
		h Key: 20			
2	Insertion	$N = 3$, [10, 30, 40], Insert 20 at Pos 2	Array: [10, 20, 30, 40]	Element inserted	Pass
3	Deletion	$N = 4$, [10, 20, 30, 40], Delete Pos 3	Array: [10, 20, 40]	Element deleted	Pass

Test Case	Option	Input Data	Expected Output	Actual Output	Status
4	Right Rotation	$N = 5$, [1, 2, 3, 4, 5], $k = 2$	Rotated: [4, 5, 1, 2, 3]	Matched rotation output	Pass
5	Exit	Choice 4	"Exiting program.."	Program exited	Pass

4. Source Code Listing

```
#include <stdio.h>

void print_array(int arr[], int n) {
    printf("[ ");
    for (int i = 0; i < n; i++) printf("%d ", arr[i]);
    printf("]\n");
}

void bubble_sort_and_binary_search(void) {
    int n, key;
    printf("\nEnter number of elements: "); scanf("%d", &n);
    if (n <= 0) return;
    int arr[n];
    printf("Enter %d integers:\n", n);
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
```

```

        if (arr[j] > arr[j + 1]) {
            int temp = arr[j]; arr[j] = arr[j + 1]; arr[j + 1] = temp;
        }
    }
}
printf("Sorted Array: "); print_array(arr, n);

printf("Enter element to search: "); scanf("%d", &key);
int low = 0, high = n - 1, found = 0;
while (low <= high) {
    int mid = low + (high - low) / 2;
    if (arr[mid] == key) {
        printf("Found at sorted index %d.\n", mid);
        found = 1; break;
    }
    if (arr[mid] < key) low = mid + 1; else high = mid - 1;
}
if (!found) printf("Element not found.\n");
}

void insertion_and_deletion(void) {
    int n, arr[100], sub_choice;
    printf("\nEnter size: "); scanf("%d", &n);
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);

    printf("1. Insert 2. Delete: "); scanf("%d", &sub_choice);
    if (sub_choice == 1) {
        int val, pos;
        printf("Enter value and pos: "); scanf("%d %d", &val, &pos);
        if (pos < 1 || pos > n + 1) return;
        for (int i = n; i >= pos; i--) arr[i] = arr[i - 1];
        arr[pos - 1] = val; n++;
        print_array(arr, n);
    } else if (sub_choice == 2) {
        int pos; printf("Enter pos to delete: "); scanf("%d", &pos);
        if (pos < 1 || pos > n) return;
        for (int i = pos - 1; i < n - 1; i++) arr[i] = arr[i + 1];
        n--; print_array(arr, n);
    }
}
}

```

```

void rotate_array_right(void) {
    int n, k;
    printf("\nEnter size: "); scanf("%d", &n);
    int arr[n], rotated[n];
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
    printf("Enter k: "); scanf("%d", &k);
    k = k % n;
    for (int i = 0; i < n; i++) rotated[(i + k) % n] = arr[i];
    printf("Rotated Array: "); print_array(rotated, n);
}

int main(void) {
    int choice;
    do {
        printf("\n1. Sort & Binary Search\n2. Insert/Delete\n3. Rotate
Right\n4. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: bubble_sort_and_binary_search(); break;
            case 2: insertion_and_deletion(); break;
            case 3: rotate_array_right(); break;
            case 4: printf("Exiting...\n"); break;
        }
    } while (choice != 4);
    return 0;
}

```

5. Execution & Verification Screenshots

```
kali@kali:~/HDOOC_Day30
Session Actions Edit View Help
--(kali@kali)~--
$ mkdir HDOOC_Day30
$ cd HDOOC_Day30
--(kali@kali)~/HDOOC_Day30
$ nano day30.c
--(kali@kali)~/HDOOC_Day30
$ gcc day30.c -o day30
--(kali@kali)~/HDOOC_Day30
$ ./day30

===== ARRAY OPERATIONS MENU =====
1. Sort using Bubble Sort and perform Binary Search
2. Perform Insertion and Deletion in a given valid position
3. Rotate array elements to the right by k positions
4. Exit
Enter choice (1-4): 1

--- 1. Bubble Sort & Binary Search ---
Enter number of elements in array: 5
Enter 5 integers:
Element 1: 40
Element 2: 10
Element 3: 50
Element 4: 20
Element 5: 30
Sorted Array (Bubble Sort): [ 10 20 30 40 50 ]
Enter element to search for using Binary Search: 20
Element 20 found at sorted index 1 (position 2).

===== ARRAY OPERATIONS MENU =====
1. Sort using Bubble Sort and perform Binary Search
2. Perform Insertion and Deletion in a given valid position
3. Rotate array elements to the right by k positions
4. Exit
Enter choice (1-4): 2

--- 2. Insertion and Deletion ---
Enter initial number of elements (max 100): 99
Enter 99 elements:
Element 1: 2
Element 2: █
```